

Generation, retrieval-augmented systems, and where this leaves you

COCO and the Part V corpora

LECTURE 24

GÉRON CH. 15–16

Applications of Machine Learning

BSc Mathematics of Artificial Intelligence · Third year

Where we are

Lecture 23 built a catalogue you can search with a sentence. Two weak spots remain, and both are about entries rather than about the model:

- Some entries have **no usable description** — nothing for the text tower to encode
- Some queries are **ambiguous**, and a ranking is a poor answer to a question the user has not finished asking

BOTH ARE REPAIRED BY GENERATION

A multimodal model can *write* the missing description. And a generator with the retriever attached can answer the ambiguous query in words, grounded in what the catalogue actually contains.

Today

1. Captioning: filling the holes in the catalogue (20 min)
2. Retrieval-augmented generation — Lecture 20's retriever joined to Lecture 18's generator (25 min)
3. Failures that belong to the join rather than to either half (15 min)
4. **The course, as one argument** (25 min)

THE FIRST REPAIR

Writing the missing descriptions

20 minutes

The hole in the index

60 of 200 catalogue entries have no description. A text index cannot rank a document that does not exist, so those entries score exactly 0% — structurally, not statistically.

THE IDEA

We have a model that maps images to text: Lecture 23 measured it running in the image-to-text direction. Use a model built for *generating* the text rather than retrieving it, and write the missing descriptions.

The captioner

Component	What it is
Image encoder	a vision transformer — Chapter 16
Text decoder	a transformer decoder with cross-attention to the image — Chapter 15
Training	image–caption pairs, next-token prediction
Size	224 million parameters

Architecturally this is the encoder–decoder of Chapter 15 with an image encoder in place of the text one. Nothing new; a different pair of modalities in the same box.

The repair, in nine lines

```
1  from transformers import BlipForConditionalGeneration, BlipProcessor
2
3  proc = BlipProcessor.from_pretrained("Salesforce/blip-image-captioning-base")
4  cap  = BlipForConditionalGeneration.from_pretrained(
5      "Salesforce/blip-image-captioning-base").to(device).eval()
6
7  with torch.no_grad():
8      batch = proc(images=missing_images, return_tensors="pt").to(device)
9      ids   = cap.generate(**batch, max_new_tokens=30, num_beams=3)
10
11  generated = proc.batch_decode(ids, skip_special_tokens=True)
12  assert len(generated) == len(missing_images)
```

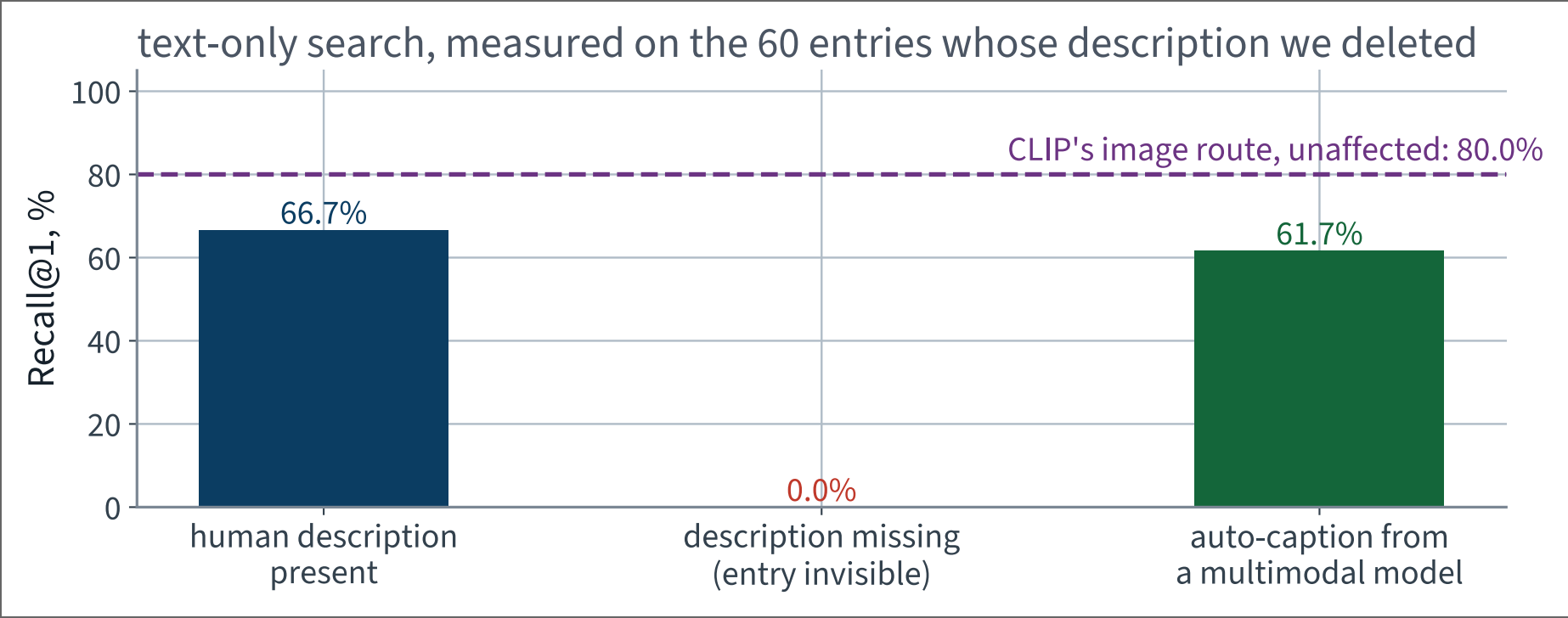
Expected wall clock: 21 s for 60 images with beam search. Not instant; run it offline, once, per new entry.

What it writes

Human description (deleted)	Generated description
This wire metal rack holds several pairs of shoes and sandals	a dog laying on top of a pair of shoes
A green vase filed with red roses sitting on top of table.	a vase of flowers sitting on a window sie
A giraffe bending over while standing on green grass.	the gi is brown and white

The first row is the interesting one: **both sentences are true**. There is a dog asleep on the shoe rack, and the two writers chose different subjects. The third row is not a sentence at all.

Better than nothing, measured



From 0% to 61.7% on the 60 affected entries — against 66.7% if a person had written them.

The whole catalogue, three ways

Text-only route, all 200 queries	R@1	R@10
Every entry described by a person	61.5%	96.0%
X 60 entries with nothing	X 46.0%	X 68.5%
✓ Those 60 auto-captioned	✓ 62.0%	✓ 94.0%

At rank 1 the recovery is complete to within one query out of 200 — which is noise, not a win over the human writers. At rank 10 it is not complete, and that gap is the real finding.

Meanwhile, the route that never read a description

On the same 60 entries	R@1
Text index, description deleted	X 0%
Text index, auto-captioned	61.7%
✓ Joint-space image route	✓ 80.0%

THE DESIGN LESSON

Two routes that fail for different reasons are worth more than one better route. The image route is immune to missing text; the text route is immune to a photograph that does not show the distinguishing feature.

Three things not to claim

- A generated caption is **not evidence about the product**. It is a restatement of the photograph, and it will confidently name things that are not there
- The recovery is measured on 60 entries of 200. That is a small sample and the interval around it is wide
- We are scoring generated captions against human captions of the same image — a friendly test. A customer's wording is not a COCO caption

THE FAILURE MODE TO WATCH FOR

An auto-caption that is wrong makes an entry *findable under the wrong query*, which is worse than unfindable. Nothing we measured today detects that.

Captioning: what it is, and is not

A multimodal model conditioned on an image, generating text. Architecturally it is Lecture 18's decoder with cross-attention into Lecture 23's image encoder — nothing new, arranged differently.

WHAT A GENERATED CAPTION IS

A **plausible** description, not a true one. It describes what the model expects to see, and it will confidently name objects that are not there. For catalogue entries that had *nothing*, plausible is an improvement on absent; as a replacement for a real description it is not.

So the honest use is narrow and worth stating: fill holes, mark the filled entries as machine-written, and measure retrieval with and without them.

Prompting is a hyperparameter

A generator's output depends on how you ask, and the difference is often larger than the difference between models.

WHICH MAKES IT A CHOICE, WITH ALL THAT IMPLIES

Trying five prompts and keeping the one with the best test score is choosing a hyperparameter on the test set — Lecture 2's error, in a costume that does not look like modelling at all. It has a validation set like anything else.

X Report the prompt with the result. A number without the prompt that produced it is not reproducible, and prompts are rarely written down because they do not feel like code.

What generation does not fix

- **It does not add knowledge.** A caption is a guess from pixels; RAG is a lookup. Neither creates information the system does not have
- **It does not calibrate.** Fluent and confident is the default register regardless of whether the answer is grounded
- **It does not remove the need for evaluation.** It makes it harder: the output is text, and text has no obvious metric

That last point is the honest closing note of the technical half of this course. Every lecture until now had a number. Here you have to *construct* the number first, and constructing it well is most of the work.

The verdict

For the first time in twelve applications, the number you committed to was probably *too low* rather than too high — because the failure this time was not in the evaluation. It was in the geometry, and the repair was somebody else's pretraining run.

STILL WORTH CHECKING

72.5% over 200 candidates is not 72.5% over fifty thousand. Recall rises as the haystack shrinks, and a catalogue is a haystack that grows.

Two things we measured and did not fix

#	Measured last lecture	Number
1	Text search R@1 on entries with no description	X 0%
2	Ambiguous queries with a single right answer	X none exist

The first is a hole in the index. The second is a mismatch between what the customer asked and what a ranked list can say.

Both are repaired in the next thirty minutes, and both are re-measured before we leave.

THE SECOND REPAIR

Retrieval-augmented generation

25 minutes · examinable

The join, drawn

1. The user asks a question in words
2. The **retriever** of Lecture 20 embeds it and returns the k most similar catalogue entries
3. Those entries are placed in the **generator**'s context, with the question
4. The generator answers *from what it was given*, and cites which entry it used

WHY THIS IS NOT JUST A BIGGER MODEL

The knowledge lives in the index, not in the weights. Add a product and it is answerable immediately; remove one and it stops being cited. No retraining, and the system can say *where* an answer came from — which is usually the requirement that decides whether it can be deployed at all.

What each half contributes

	Good at	Bad at
Retriever	finding the right entries, fast, over millions	saying anything; it only ranks
Generator	fluent answers, synthesis across entries	knowing anything reliably; it will produce a plausible answer regardless

Each covers the other's weakness, which is why the combination is worth more than either. It is also why the failures are interesting: they belong to **neither half**.

Failures of the join

THREE, AND NONE IS VISIBLE FROM EITHER COMPONENT

Retrieved but ignored. The right entry is in the context and the generator answers from its own weights instead.

Not retrieved, answered anyway. The retriever misses, and the generator does not say so — it produces a fluent answer with nothing behind it.

Retrieved and misread. The entry is present and correct, and the answer misstates it.

X Evaluating the retriever alone (recall@k) and the generator alone (fluency) can both look excellent while the system does all three. The join needs its own measurement.

Chunking, and why it decides everything

Before anything is retrieved, the corpus must be cut into units. That choice is upstream of the retriever and of the generator, and it is usually made without measurement.

- **Too small** — a chunk lacks the context that makes it answerable, and the generator gets a fragment
- **Too large** — the embedding averages several topics and matches none of them well
- **Split badly** — a sentence cut in half retrieves for neither of its halves

X It is a hyperparameter. Treat it as one: vary it, hold everything else fixed, and measure recall@k on the same judgements — Lecture 19's protocol, unchanged.

How many to retrieve

k trades two failures against each other.

Small k

the right entry may not be there at all

the generator cannot recover

Large k

it is there, buried among distractors

the generator may attend to the wrong one

WHY RE-RANKING EARNS ITS COST HERE

Retrieve a generous k with the cheap bi-encoder, then re-rank with the cross-encoder of Lecture 20 and pass only the top few on. High recall from the first stage, high precision from the second — which is the two-stage argument of Part V, arriving where it matters most.

How to measure the join

- **Groundedness** — is every claim in the answer supported by a retrieved entry? Check mechanically where you can
- **Attribution** — does the cited entry actually say it? A citation that does not support its claim is worse than none
- **Abstention** — when nothing relevant is retrieved, does it decline? Measure this deliberately, with queries whose answer is genuinely absent
- **End to end** — and only then, is the answer right?

THE CLOSED-BOOK CONTROL

Run the same questions with the retrieval turned off. Whatever the system still answers correctly, it knew already — and the difference is what retrieval actually bought. Same shape as every ablation in this course.

What the customer actually types

- “something to sit on”
- “gear for bad weather”
- “an animal that would suit a small flat”

A ranked list of pictures is a poor answer to any of these. The customer wants a shortlist *with reasons* — and a ranking has no place to put a reason.

Twelve such queries, written by hand and listed in the notebook. There is no dataset of ambiguous queries; inventing them and saying so is the honest option.

Two options that do not work

OPTION A

Show the top ten pictures and let the customer decide. That is what we already have, and it is why they asked in words.

OPTION B

Ask a language model. It has never seen our catalogue, so anything it recommends is invented — and it will invent it fluently.

Retrieval-augmented generation is the composition of the two: **retrieve first, then generate strictly from what was retrieved.**

First, decide how you will check it

“The answer is good” is not measurable in a lecture. So give every entry a stock number and require the assistant to cite them:

THE METRIC

A cited SKU either exists in the catalogue or it does not. Count the citations, count the ones that exist, report the fraction. No judgement call anywhere in the loop.

This measures *grounding*, not helpfulness. Say which one you measured; they are not the same thing and only one of them is cheap.

Retrieve, then generate

```
1  q = unit(clip_text(["something to sit on"]))          # (1, 512)
2  top5 = np.argsort(-(q @ I.T)[0])[:5]                 # I: catalogue images
3
4  shortlist = "\n".join(f"- {cat[j]['sku']}: {cat[j]['captions'][0]}"
5                        for j in top5)
6
7  prompt = (f'Here are the catalogue entries our search returned for '
8            f'"something to sit on":\n{shortlist}\n\n'
9            f'Recommend the best ones, citing only SKUs from the list above. '
10           f'If none fit, say so.')
11
12 answer = generate(prompt)                             # a 0.5B instruction model
13 cited  = re.findall(r"CAT-\d{6}", answer)
14 print(f"{sum(s in valid_skus for s in cited)}/{len(cited)} cited SKUs exist")
```

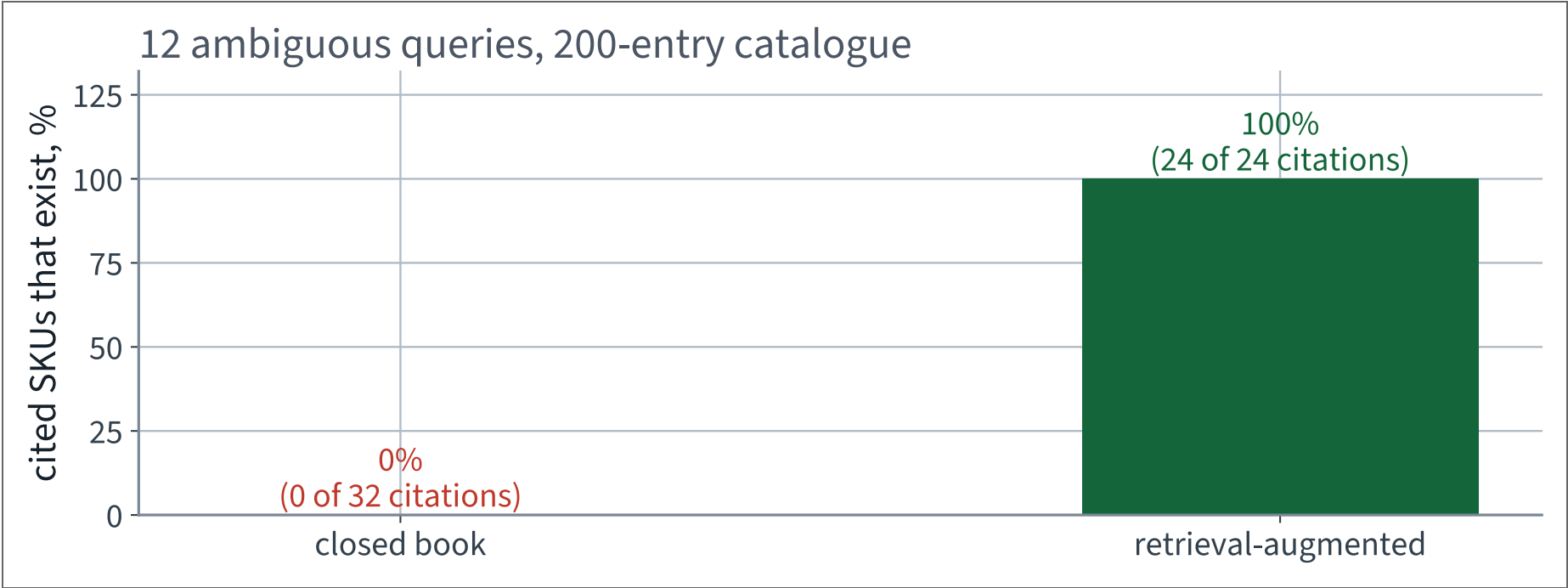
Closed book — what it does without retrieval

ASKED THE SAME QUESTION WITH NO CATALOGUE

The model produces a fluent recommendation and cites stock numbers in exactly the right format. Across 12 queries it emitted 32 citations, of which 0 correspond to an entry that exists.

The failure is not that it refuses. It is that it does not refuse, and nothing in the output distinguishes an invented SKU from a real one.

With the catalogue in the prompt



0% of closed-book citations exist against 100% when the shortlist is in the prompt — over 12 queries and a 200-entry catalogue.

The numbers

Over 12 ambiguous queries	Closed book	Retrieval-augmented
Queries that cited at least one SKU	10	12
Citations emitted	32	24
Citations that exist	X 0	✓ 24
Grounding rate	X 0%	✓ 100%

The generator is a 494-million parameter instruction model, chosen so the whole thing runs on a laptop. A larger one would follow the citation instruction more reliably; it would not change the shape of the result.

What this does **X not** fix

- **Grounding is not correctness.** A cited entry can exist and still be a bad recommendation. We measured the cheap half
- **The retriever is now the ceiling.** If the right entry is not in the top five, no amount of generation will recover it — which is why the R@5 we measured last lecture is the number that matters here
- **Fluency is unchanged.** A wrong answer built from real SKUs reads better than a wrong answer built from invented ones

WHERE THIS SITS IN THE FIELD

Evaluating generated answers — faithfulness, attribution, judging with another model — is a live research area and entirely outside Chapters 1–16. Not examinable, and worth knowing exists.

Everything, re-measured

Route	Before	After
Text query → image, joint space	72.5%	72.5% (untouched)
Text index, on the entries with no description	X 0%	✓ 61.7%
Ambiguous queries, grounding rate	X 0%	✓ 100%

Two repairs, two measured deltas, one number deliberately unchanged as a control. That last row is what makes the other two believable.

The notebook for this lecture

Open Lecture 24 in Colab

- **Runs on free CPU.** Small pretrained models, inference only
- Captions generated for the entries that have none, and the retrieval re-measured with them
- The closed-book control: the same questions with retrieval off, so what retrieval bought is a number

WHAT TO CHANGE

Ask a question whose answer is *not* in the catalogue. A system that answers it anyway has the second failure of the join, and you have just built the test that finds it.

THE LAST TWENTY-FIVE MINUTES

**The course,
as one argument**

25 minutes

The argument

Every method in this course arrived because something measurable could not be done without it.

AND EVERY LECTURE HAD THE SAME SHAPE

A quantity that has to be derived before the method makes sense; the method; a worked example with real numbers; and the conditions under which it fails. Eighteen derivations, and they are the part that will still be true when every library in these notebooks has been replaced.

What the first lecture promised

SLIDE 2, LECTURE 1

“Twelve real applications, each built until it breaks, then repaired with the method that fixes it. No chapter introduces itself. Every technique arrives because something *you measured* went wrong without it.”

That was a claim, not a plan. Here is the evidence.

The claim, stated precisely

Every method you met in this course was introduced at the moment a number *you produced* was wrong without it.

Not “was useful”. Not “is standard”. **Was wrong, and measurably so, and you were the one holding the wrong number.**

The rest of this block is that claim, twelve times, with the measurement attached each time.

Twelve applications, twelve first failures

Twelve applications, twelve things that broke first

the red line is the measurement that failed; the method arrived afterwards

1–2 · housing prices
scaled before splitting,
scored on the training set
→ pipelines, cross-validation

3–4 · rare-event detector
96.9% accurate, and it
missed 22.8% of the fives
→ precision, recall, thresholds

5–6 · survival probabilities
degree 5 fit the training
set and nothing else
→ bias–variance, ridge, lasso

7–8 · land cover
one tree, refit, gives a
different explanation
→ bagging, forests, extra-trees

9–10 · face grouping
4,096 dimensions, and a
sweep that took a minute
→ SVD, PCA, projection

11–12 · document sorting
a loop you cannot open,
and two silent bugs
→ PyTorch, autograd, eval()

13–14 · deep recognition
twenty layers, and the
signal died on the way
→ initialisation, normalisation

15–16 · visual inspection
from scratch, on far too
few labelled images
→ transfer learning, freezing

17–18 · object detection
a visibly wrong box, and
no number for how wrong
→ IoU, GloU, average precision

19–20 · demand forecasting
a random split across
time: you forecast the past
→ stationarity, backtesting

21–22 · customer feedback
embeddings learned from
one small labelled corpus
→ pretraining, fine-tuning

23–24 · the catalogue
two encoders, two spaces,
and nothing between them
→ the contrastive objective

Read the middle line of each box. That is the syllabus — the chapter order is a consequence of it.

Part I — the ideas everything else reuses

Lecture	The derivation	Where it came back
2 · end-to-end	least squares, the normal equation	ridge in 5; collinearity everywhere
3 · classification	imbalance; precision is not monotone	the maximum inside AP, Lecture 14
4 · training models	gradient descent	every network from Lecture 9 on
5 · regularisation	bias–variance	every learning curve since
6, 7 · trees, ensembles	impurity; variance of a correlated average	multi-head attention, Lecture 18
8 · PCA	SVD; Johnson–Lindenstrauss	matrix factorisation in 21; the sphere in 23

Parts II–IV — the same three ideas, deeper

Lecture	The derivation	Where it came back
9 · networks	what a layer computes	everything after it
10 · PyTorch	reverse-mode autodiff	why training is affordable at all
11 · deep training	variance propagation	BPTT in 16; residuals in 18
12 · convolution	weight sharing, equivariance, memory	what a ViT gives up, Lecture 23
14 · detection	IoU; mAP	ranking metrics in 19
15 · time series	stationarity; autocorrelation	causal masking in 18
17 · text	softmax, cross-entropy, logits	attention's softmax in 18; the contrastive loss in 23

Part V — where the embeddings earned their living

Lecture	The derivation	The point
19 · lexical IR	MRR, AP, NDCG	a ranking is not a classification, and needs its own metrics
20 · dense IR	—	two encoders and an inner product; the negatives are most of the work
21 · matrix factorisation	low rank, and its relation to the SVD	Lecture 8, with missing entries
22 · neural recsys	—	the two-tower model IS the bi-encoder; and offline evaluation is where recsys is usually wrong
23 · multimodal	contrastive objective, temperature	the same two towers, different modalities

Lectures 20, 22 and 23 are three uses of one object. That is the strongest single thread in the course, and it is why Part V sits after transformers rather than before.

The eighteen derivations, in one place

- Least squares and the normal equation · 2
- Imbalance; precision is not monotone · 3
- Gradient descent · 4
- The bias–variance decomposition · 5
- Impurity: Gini and entropy · 6
- Variance of a correlated average · 7
- PCA via the SVD; Johnson–Lindenstrauss · 8
- What a layer computes · 9
- Reverse-mode automatic differentiation · 10
- Variance propagation and initialisation · 11
- Weight sharing, equivariance, memory · 12
- IoU’s vanishing gradient; mAP · 14
- Stationarity, differencing, autocorrelation · 15
- Softmax, cross-entropy and logits · 17

• Scaled dot-product attention · 18

• Evaluating a ranking: MRR, AP, NDCG · 19

• The failures, collected

• Matrix factorisation and the SVD · 21

• The contrastive objective and its temperature · 22

Failure	First met	Measured cost
Fitting before splitting	Lecture 2	about a dollar — and unknowable in advance
Evaluating on training data	Lecture 2	an RMSE of zero
Accuracy under imbalance	Lecture 3	a useless model above 90%
Choosing on the test set	Lectures 2, 5	an optimistic report, every time
Missing <code>zero_grad()</code>	Lecture 10	76.7 points
Default initialisation, deep	Lecture 11	a network that does not train at all
Random split on a series	Lecture 15	optimistic, and undeployable
Augmenting the validation set	Lecture 13	1.57 points of noise on the criterion

X Not one of them raises an exception. That is the whole reason the course is organised around measurement.

Application 9, in numbers

Detection on 128 COCO images	Value
Objects miscounted per image, at the default cut-off	X 3.0
...at the cut-off chosen for the task	1.92
mAP at IoU 0.5	0.659
mAP averaged over IoU 0.5 to 0.95	0.439
X The same detections, averaged per image	X 0.817

The last two rows are the same predictions. Averaging over images instead of over classes moves the headline by 0.157 — a mean of a mean is not a mean.

Application 10, in numbers

Same model, same rows, different protocol	MAE
Random cross-validation split	44,925 X
Honest, time-ordered split	56,446 ✓
X Optimism from shuffling time	X 25.6%

Nothing was fitted on the test set. No preprocessing crossed the split. The leak was the *ordering*, and with positive autocorrelation a point’s own near-future was in its training set.

Six failures, and where they live outside this course

Failure	Met in	Where it recurs
Leakage from fitting before splitting	L2	any target encoding, oversampling, feature selection or imputation done on the full frame
Accuracy under imbalance	L4	fraud, medical screening, click prediction, defect detection — anywhere the interesting class is rare
Variance mistaken for signal	L6, L8	a single-seed comparison; a leaderboard whose top ten are within noise of each other
Signal vanishing with depth	L14	any deep stack without residual connections or normalisation, including very long sequences
The wrong metric	L4, L18, L23	a ranked list scored with accuracy; a detector scored without IoU; a recall quoted without its candidate set

Temporal leakage

L20

any user-level split where the same user appears in both halves;
any shuffled panel data

The catalogue of silent failures, closed

It runs, and produces a plausible number	Diagnosed in
Scaler fitted before the split	L2
Evaluating on training data	L2
Accuracy under class imbalance	L4
Hyperparameters chosen on the test set	L2, L6, and again on a prompt template in L23
Missing <code>model.eval()</code> / <code>optimizer.zero_grad()</code>	L12
Metric averaged per batch, not over the set	L12
Random split on an autocorrelated series	L20
Embeddings compared without normalising	L24

X Every row of that table runs to completion. Not one of them raises.

What did not change, in twenty-four lectures

1. **Split before anything is fitted.** Lecture 2, and every lecture since
2. **Compute the trivial baseline first.** A metric with nothing beside it is decoration
3. **Two numbers measured on different rows are not comparable.** However similar the column headings look
4. **A difference smaller than the spread is not a difference.** Report the spread
5. **Change one thing.** Otherwise the difference is not attributable to anything

X Five rules, no mathematics, and they would have caught every failure in the table two slides ago.

The shape of a lecture, and why

Block	Why it is there
Where we are	so a lecture you missed is recoverable
The derivation	because the method does not make sense without it, and it outlasts the libraries
The method, with a worked example	real numbers from real data, reproducible from the repository
Failure conditions	stated as properties, never sprung as traps
The notebook	complete, correct, and printing every figure the slides state

Every number on every slide of this course is produced by a script in the repository, and every figure a slide states is one its own notebook prints. Both are checked mechanically.

The datasets, and why each

California housing	regression, and a target with a cap in it
MNIST	a rare event carved out of a balanced corpus
Titanic	small, mixed types, and a real need for probabilities
CoverType	large enough that trees and ensembles differ
Olivetti	more dimensions than samples
Fashion-MNIST, CIFAR-10	networks, then deep networks
Flowers102	too few labels to train from scratch
COCO	several objects per image, and captions
Chicago transit	ordered rows, and strong seasonality
IMDb	free text

BEING EXPLICIT

What this course did not cover

and where those gaps sit

Why this slide exists

Because “I took a machine learning course” is read by other people as a claim about what you can do. You should be able to say precisely what it does and does not mean.

THE SCOPE, ONE LAST TIME

Chapters 1–16 of one book, thoroughly, on twelve applications. That is a large and useful slice of applied machine learning. It is not the field.

Gaps: what this course did not cover

- **Reinforcement learning** — an agent acting, and learning from the consequences. A different problem shape entirely
 - **Generative image models** — diffusion, and what it optimises
 - **Causality** — every method here predicts; none of them tells you what happens if you *intervene*
 - **Fairness and model governance** — who a model is worse for, and who is accountable when it is
 - **Serving and drift** — the model in production, and how it decays quietly once the world moves
- Named so you know they exist and roughly where they sit. None is examinable, and each is a course.

The two you are most likely to need first

Drift

Every number in this course was measured on data drawn like the training data. In production that stops being true, gradually and without warning, and the score you reported at launch is the last honest one anybody has. Monitor the inputs, not just the outputs: the distribution moves before the accuracy does.

Who it is worse for

Lecture 2 sliced the error by district and found the poorest tenth predicted worst. That slice was one line of code, and it is the difference between a system you can defend and one you cannot. The habit generalises: always ask for whom the average is wrong.

Gaps that are in the book, past our chapter

Topic	Where it sits
Autoencoders, GANs, diffusion models	generation rather than prediction; the same encoders, a different objective
Reinforcement learning	learning from a reward signal instead of labelled examples — a different problem class entirely
Training and serving at scale	data pipelines, distributed training, deployment

Nothing above is examinable. All three are the natural next chapters of the same book, and you now have the vocabulary for them.

Gaps that are beyond the book entirely

Topic	Why it matters, in one line
Causal inference	every model here answers “what is associated with what”; deciding an <i>intervention</i> needs different assumptions and different data
Uncertainty quantification	we produced one confidence interval, in Lecture 2. Calibrated predictive uncertainty is its own subject
Fairness and harm	we ran one fairness check. Who a system fails, and what that costs them, is not a metric you can pick off a list
Privacy	differential privacy, federated learning, and what a model memorises about its training data
Monitoring and drift	a model is fitted once and lives for years. Detecting that the world moved is a permanent job

Experimental design an offline metric improving is not a business improving. That is an A/B test, and it is a statistics problem

And the part that moves fastest

- **Pretraining at scale.** We used four pretrained models today and trained none of them. What it takes to make one — the data, the cost, the failure modes — is not in this course
- **Instruction tuning and alignment.** The generator in the last section behaves the way it does because of a training stage we never discussed
- **Evaluating generated text.** We measured grounding because grounding is checkable. Measuring whether an answer is *good* is unsolved

THE TRANSFERABLE PART

Every one of those still fails in the ways this course taught you to look for: the wrong metric, a missing baseline, a test set that leaked, and a number quoted from one run.

What you can measure without a ground truth

- **Consistency** — ask the same thing twice, in two wordings. Disagreement bounds reliability from above
- **Groundedness against the retrieved text** — mechanical, and it does not need a gold answer
- **Abstention on unanswerable queries** — construct them deliberately; this is the cheapest useful test of a RAG system
- **Human judgement on a sample** — expensive, and still the anchor everything else is validated against

THE RULE, UNCHANGED SINCE LECTURE 1

A metric with nothing to compare it to is decoration. That is as true of a generated answer as it was of an RMSE, and harder to honour.

The five questions, for the last time

1. What touched the test set?
2. What was fitted, and on what?
3. What is the shape here?
4. What was dropped — rows, columns, NaNs? Count them
5. What is the default I did not ask for?

Question 3 caught a crash in Lecture 23. Question 5 caught two assistant failures in two consecutive lectures. They keep working.

The four rules, unchanged since the first lecture

1. You may not submit code you cannot explain line by line
2. Every number needs a baseline
3. Every model needs an ablation
4. Report the failure cases

Rule 2 did the most work this application: the difference between “cosine 0.006” and “cosine 0.006, against 0.003 for unrelated pairs” is the whole of Lecture 23.

What you can do now

1. Take a problem, frame it, and say what a correct answer would be *before* fitting anything
2. Choose a metric that matches the brief, and measure the trivial baseline first
3. Build the simplest thing that runs, inside a pipeline, evaluated on a protocol you fixed in advance
4. Read the failure: slice the error, look at the worst cases, and name the conditions under which the method breaks
5. Improve one thing at a time, and report the ablation rather than the best row
6. Write down what the number was measured on, every time

Six steps. Nothing in them is specific to a library, and none of them will be out of date in ten years.

A last word on the assistant

You will build most of this with an assistant, and Lecture 1 set out how: specify, generate, *read*, test, verify — and steps three to five are yours.

WHAT HAS CHANGED BY LECTURE 24

You now know what to look for. “What was this measured on?”, “what is the baseline?”, “was anything fitted before the split?” — these are the questions an assistant will not ask itself, and they are precisely the ones that decide whether the number it hands you means anything.

X The code was never the scarce part. Knowing whether to believe the output is, and that is what these twenty-four lectures were for.

The examination

Written — two hours, closed book

- A · the derivations (40%)
- B · choosing a method for a stated situation (35%)
- C · reading results (25%)

Oral — twenty minutes

- Any topic from the course
- Including a cell from one of these notebooks
- No notes, no computer

18/30 on each, independently; the final mark is the average. Everything in Chapters 1–16 and in the Part V notes is examinable; nothing marked *engineering* or *for context* is.

How to keep learning

- **Reproduce something.** Take a paper's headline number and try to get it. You will learn more from the gap than from the paper
 - **Keep the habit of the baseline.** It is the single cheapest defence against a wrong conclusion, and almost nobody does it
 - **Read the evaluation section first.** If it does not say what was held out and how, the rest is decoration
 - **Write the check before the model.** Every notebook in this course does, and it is why they still run
- The libraries in these notebooks will be superseded. The derivations will not, and neither will the habit of asking what a number was measured on.

Every notebook in the course

01 • What machine learning is, and how we will work

02 • The end-to-end project

03 • Classification and its metrics

04 • Training models

05 • Regularisation and the bias–variance trade-off

06 • Decision trees

07 • Ensembles and random forests

08 • Dimensionality reduction and unsupervised learning

09 • Neural networks, from the perceptron up

10 • PyTorch

11 • Training deep networks

12 • Convolutional networks

13 • Transfer learning

14 • Detection and segmentation

15 • Time series

16 • Recurrent networks

17 • Text

18 • Attention and transformers

19 • Information retrieval: the lexical foundation

20 • Information retrieval: dense retrieval

21 • Recommender systems: from ratings to factors

22 • Recommender systems: neural, and evaluated honestly

23 • Vision transformers and multimodal retrieval

24 • Generation, retrieval-augmented systems, and where this leaves you

The full index of all twenty-four is on the course site. Every one of them is ours, written from a specification, in the lecture.

In the book

Topic	Chapter
Softmax, cross-entropy, logits	4, 15
Transformer encoders and decoders; cross-attention	15
Vision transformers; text–image models; zero-shot	16
Image captioning	16
Random projection and concentration	7

Retrieval-augmented generation, the modality gap, and the evaluation of generated answers are outside Chapters 1–16. They are here because the application needed them, and they are flagged as non-examinable.

What we did today

1. Took the contrastive objective apart: the sphere, the target of zero, the temperature, the negatives
2. Measured the concentration result on 20,000 random pairs and on our own embeddings, and found the modality gap
3. Read the learned temperature off the checkpoint: 0.01000
4. Watched the same loss lose 21 points of top-1 accuracy from one missing normalisation
5. Filled 60 missing descriptions automatically and recovered R@1 from 0% to 61.7% on those entries
6. Grounded an assistant in the catalogue and moved its citation validity from 0% to 100%

If you remember one thing

THE LINE

A number is only as good as the question “measured on what?” can be answered.

Every failure in this course — the leak, the training score, accuracy under imbalance, the random split on a series, the augmented validation set, the ungrounded generated answer — is a case where that question had an answer nobody had asked for.

Asking it costs seconds. Not asking it has cost, in these twenty-four lectures alone, an RMSE of zero, a 90% classifier that never fires, a network that does not train, and a forecast of the past.

NOT PRESENTED — FOR AFTERWARDS

Exercises

Lecture 24

five, in the style of the written paper

Exercises — Lecture 24

- 1** A contrastive loss is written as `img @ txt.T / tau` on the output of `get_image_features`. State the defect, and the assertion that would have caught it. [5]
- 2** An auto-caption is generated for catalogue entries that have no description. State what it recovers, and the failure mode nothing in the evaluation detects. [5]
- 3** A retrieval-augmented system is measured by whether cited stock numbers exist. State what that measures, and what it does not. [4]

Solutions overleaf — there is no Lecture 25.

Exercises — Lecture 24 (continued)

- 4** In a retrieval-augmented pipeline the retriever's recall bounds the whole system. Explain why, and what that implies about where to invest. [4]
- 5** Give the five rules the course ends on, and name the one that would have prevented the largest number of the failures it measured. [5]

Solutions overleaf — there is no Lecture 25.

THE FIVE SET AT THE END OF LECTURE 23

Solutions

Lecture 23

not part of this lecture

Solutions — Lecture 23

1. Two encoders are trained separately, one on images and one on text. Explain why their cosine similarity...
✓ Any rotation of the image space leaves every image–image cosine and the image loss unchanged, while changing every image–text cosine.
- The objective cannot distinguish the rotated space from the original
- So it did not pick one, and the relative geometry is arbitrary
2. In d dimensions two independent random unit vectors have a cosine with standard deviation $1/\sqrt{d}$
✓ Near zero, with a spread of about 0.044. There is exactly one point at cosine -1 , so most pairs cannot be there.
- Unrelated means orthogonal in high dimensions, not opposite
- It is why a contrastive loss targets zero for a non-matching pair

Solutions — Lecture 23 (2)

3. A contrastive loss is computed at $\tau = 1$ on cosine scores. State why it barely trains, and what τ ...

✓ Cosines span at most 2, so the softmax is nearly uniform. τ changes where the gradient goes, not which column is largest.

- A range of 2 gives a ratio of at most e^2 across all competitors
- Temperature cannot change the accuracy of a fixed similarity matrix

4. Recall@10 is reported over 200 candidates. State the exact random baseline, and why it is computed rather...

✓ $10/200 = 5\%$.

- One relevant item ranked uniformly gives $P(\text{rank} \leq k) = k/n$
- A closed form needs no seed and carries no noise

Solutions — Lecture 23 (3)

5. An entry with no written description scores zero on a text retrieval route. State whether this is a ranking...

✓ **Structural — an entry with no text is not in a text index at all.**

- No amount of re-ranking reaches a document that was never indexed
- The repair is another route, not a better scorer

ANSWERED HERE, BECAUSE THERE IS NO LECTURE 25

Solutions

Lecture 24

not part of this lecture

Solutions — Lecture 24

1. A contrastive loss is written as $\text{img} @ \text{txt.T} / \text{tau}$ on the output of `get_image_features`. State the defect, and...
✓ **The features are not normalised, so the product is not a cosine. Assert every row has unit norm.**
 - The entries carry both magnitudes, so the temperature divides a quantity with no fixed scale
 - Magnitude tracks length and token frequency, so long documents win by default
2. An auto-caption is generated for catalogue entries that have no description. State what it recovers, and the...
✓ **It recovers part of the lost recall. A caption that is wrong makes an entry findable under the wrong query.**
 - Scoring generated captions against human captions of the same image is a friendly test
 - Worse than unfindable, and no metric here sees it

Solutions — Lecture 24 (2)

3. A retrieval-augmented system is measured by whether cited stock numbers exist. State what that measures, and...

✓ **It measures grounding, not helpfulness.**

- A cited entry can exist and still be a bad recommendation
- It is the cheap half, and saying which half was measured is the point

4. In a retrieval-augmented pipeline the retriever's recall bounds the whole system. Explain why, and what that...

✓ **If the right entry is not retrieved, no generation recovers it.**

- The generator can only work with what it is given
- The first stage sets the ceiling, exactly as in Lecture 20

Solutions — Lecture 24 (3)

5. Give the five rules the course ends on, and name the one that would have prevented the largest number of the...

✓ **Split before fitting; preprocessing inside the cross-validated object; nothing from the test set in the training path; fixed seeds with per-fold scores; every number gets a baseline. The baseline rule.**

- Most measured failures were numbers with nothing to be read against
- A baseline is the cheapest of the five and catches the most

Thank you

Eighteen derivations · twenty-four notebooks · thirteen datasets

Every number on every slide reproduced by a script in the repository

Questions, and then the examination